MoE-Based Customer Segmentation Model

A multimodal study of behavioral, text, and temporal customer representations

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Analytical report

Brazilian E-Commerce Public Dataset (Olist)

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Prepared for research and decision support

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1 Executive summary

This report evaluates whether a customer-specific Mixture of Experts (MoE) model can produce more useful customer segments than strong unimodal and static-fusion alternatives. The study uses a temporally defined cohort of 57,510 customers from the Olist Brazilian e-commerce dataset and combines three complementary views: structured behavioral profiles, review-text semantics, and order-history sequences. The evidence is mixed in a useful way. A simple learned gate did not improve on static fusion: the best Phase 10 routing variant reached a silhouette of 0.374525, well below the static text-plus-temporal baseline of 0.557928. After redesigning the gate as a hierarchical, cluster-aligned, sparse-routing system with entropy regularization and five-seed ensemble averaging, Phase 13 reached 0.590873 silhouette, 0.638034 Davies–Bouldin index, and 19,223.176 Calinski–Harabasz score. Thus, adaptive routing is supported conditionally: it works when the routing objective and architecture are designed to preserve cluster geometry, not when a small gate merely reconstructs an average representation.

The results also reveal an important business distinction. The strongest geometry is not automatically the strongest retention discriminator. The simple KMeans RFM baseline produced the highest corrected retention lift—3.756×, with a maximum cluster retention rate of 4.706%—and remains the most defensible choice for retention-focused activation until Phase 13 retention is evaluated directly.

Bottom line. Use the hierarchical MoE as the leading segmentation engine for multimodal structure, retain RFM as an interpretable retention benchmark, and treat availability-aware masking plus strict temporal controls as non-negotiable components of the system.

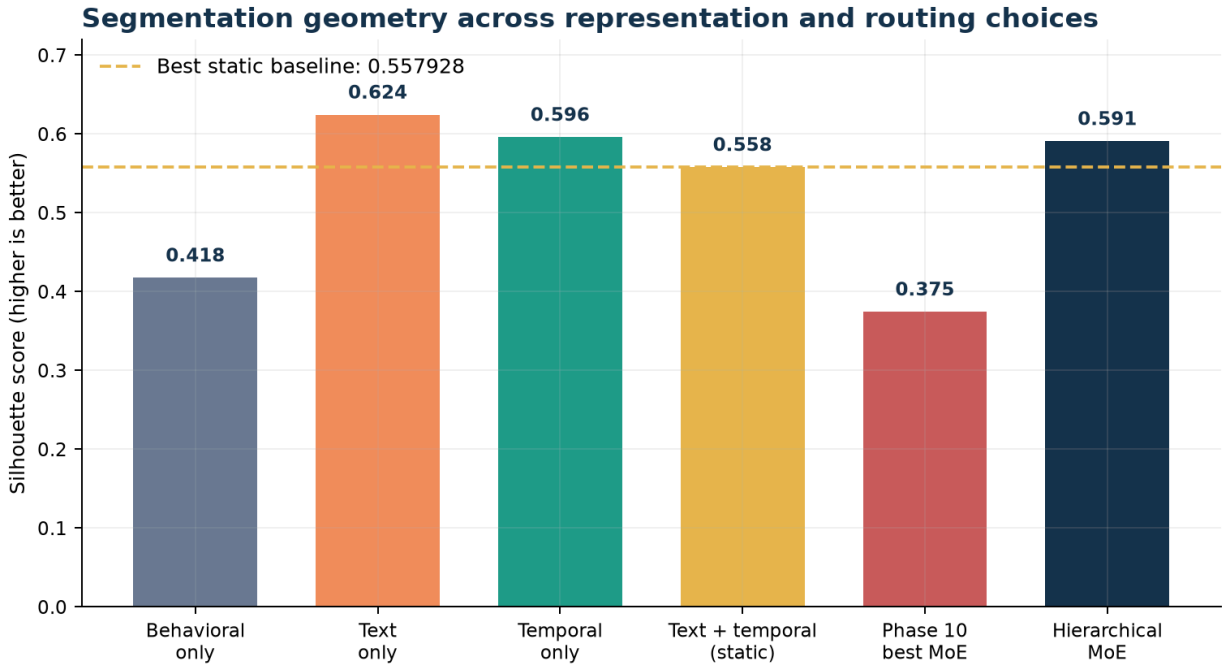


Figure 1: Silhouette comparison across representation and routing choices. The hierarchical MoE clears the strongest static-fusion baseline.

2 Research question and evaluation logic

The central question is: **Can customer-specific adaptive routing across behavioral, text, and temporal views improve segmentation over strong unimodal and static-fusion baselines?** The study answers this in stages rather than comparing a single final model against a weak benchmark.

First, legacy clustering establishes the value of simple, interpretable behavioral features. Second, independent text and temporal representations are created under an observation cutoff. Third, fixed-weight multimodal fusion provides a strong non-adaptive benchmark. Fourth, MoE routing is tested first with simple gates and then with increasingly structured routing mechanisms. Finally, downstream retention separation is assessed where the available manifests support it.

The primary geometry metric is the silhouette score, where larger values indicate better within-cluster cohesion and between-cluster separation. Davies–Bouldin (DBI) is interpreted in the opposite direction, with lower values preferred. The Calinski–Harabasz (CH) score provides a scale-sensitive separation diagnostic, while stability ARI measures assignment consistency across repeated fits. Retention is evaluated separately because a geometrically clean segmentation may not be the segmentation that best isolates future repeat purchasers.

3 Data foundation and cohort design

The underlying Olist data contains customer, order, item, payment, review, product, seller, and geolocation tables. For modeling, these are consolidated into customer-level analytical views rather than carried forward as a raw warehouse inventory. The research cohort contains 57,510 customers, split into 40,208 training customers (70%), 8,681 validation customers (15%), and 8,621 test customers (15%). The observation cutoff is 1 March 2018, followed by a six-month future evaluation window.

Cohort element	Customers	Role
Training	40,208	Fit representations and preprocessing
Validation	8,681	Select configurations and hyperparameters
Test	8,621	Blind downstream evaluation
Total	57,510	Research cohort
Test retention baseline	1.253%	108 retained customers

The views are intentionally asymmetric. Behavioral and temporal features cover all customers, while text is available for 22,535 customers (39.2%). This missingness is not a nuisance to be silently imputed away: it is part of the modeling problem. The final system therefore masks unavailable experts and renormalizes routing weights over the views that exist.

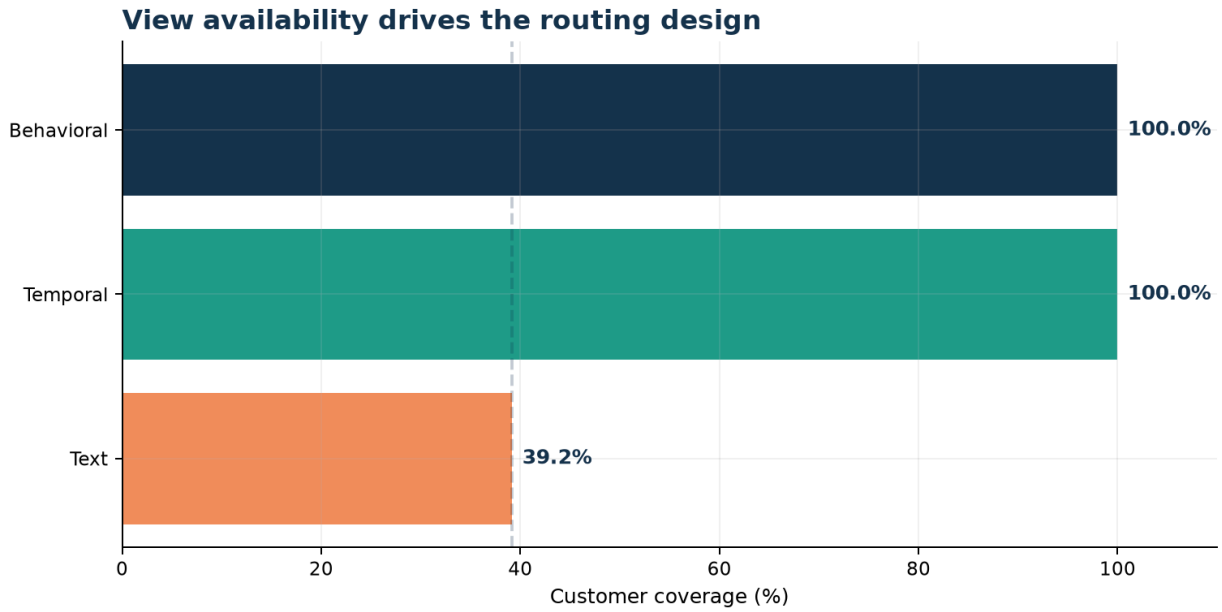


Figure 2: Customer coverage by modality. The limited text view makes explicit availability handling central to the design.

4 Representation design

4.1 Behavioral view

The behavioral profile contains 38 engineered features spanning purchase frequency, monetary value, items, categories, sellers, recency, lifetime, logistics, payment, reviews, and derived ratios. For the expert model, eight selected non-redundant features are projected to 32 dimensions using a StandardScaler and PCA fitted on training customers only. The behavioral view is the most complete and interpretable view, but its geometry is not always the strongest when used alone.

4.2 Text view

Eligible review text is embedded using `all-MiniLM-L6-v2`, producing 384-dimensional review representations that are projected to 32 dimensions. The pipeline retains 22,535 customers with text after removing duplicates and empty reviews. Critically, only reviews created before the observation cutoff are eligible; the audit found zero cutoff violations and zero non-finite customer embeddings.

4.3 Temporal view

Order histories are represented as sequences of pre-cutoff customer events and encoded with a GRU autoencoder into 32-dimensional vectors. There are 59,427 events before the cutoff, all 57,510 customers have history, and no customer is truncated by the maximum sequence length of 20. Later-known fields such as delivery duration, review scores, and final statuses are excluded from the temporal input to prevent future information from leaking backward into the representation.

View	Coverage	Model dimension	Representation
Behavioral	100%	8 → 32	Engineered profile + PCA

View	Coverage	Model dimension	Representation
Text	39.2%	384 → 32	Sentence-transformer + PCA
Temporal	100%	32	GRU autoencoder

5 Baselines: what the problem looks like before MoE

The legacy baselines establish two different notions of success. KMeans on RFM features gives the strongest retention separation and a respectable silhouette of 0.5314. Temporal-only modeling produces stronger geometry at 0.596058, while text-only modeling reaches 0.624421 but with a much weaker DBI and lower stability. These results caution against treating any single metric as a complete definition of model quality.

The strongest static multimodal baseline is fixed-weight text plus temporal fusion: silhouette 0.557928, DBI 0.779689, CH 9,113.616, and stability ARI 1.000000. Adding the behavioral view to all three modalities reduces silhouette to 0.305570, suggesting that more views do not automatically mean more useful geometry. The availability-aware indicator is important: removing availability information from text-plus-temporal fusion drops silhouette to 0.319270 and stability ARI to 0.862250.

Model	Silhouette	DBI	CH	Stability ARI
KMeans RFM	0.5314	0.6182	7,213.430	0.9937
Temporal-only	0.596058	0.663542	18,176.656	1.000000
Text + temporal, fixed weight	0.557928	0.779689	9,113.616	1.000000
All three, fixed weight	0.305570	1.245361	6,564.488	0.998652
Indicator-free text + temporal	0.319270	1.270539	5,502.779	0.862250

6 MoE experiments and the routing lesson

6.1 Phase 10: simple gates are not enough

The first MoE implementation tested uniform routing, a fixed behavioral-heavy route, a learned gate, and a quality-aware gate. None approached the static text-plus-temporal baseline. The best Phase 10 result was the fixed behavioral-heavy route at 0.374525 silhouette, a deficit of 0.183403 versus static fusion. The learned gate reached only 0.268705 and had lower stability ARI (0.978325).

This failure is informative. The gate optimized reconstruction of an average projected representation, not cluster separation itself. A gate can therefore produce plausible mixture weights while moving customer representations toward a less separable compromise. The negative result rules out the simpler claim that customer-specific routing is automatically beneficial.

6.2 Phase 13: redesign around cluster geometry

The successful model changes both the architecture and the optimization target. A coarse gate first chooses between behavioral and the text-plus-temporal branch; a fine gate then chooses between text and temporal within that branch. Both gates use quality-aware features, a 128-unit LayerNorm/GELU hidden layer, DEC-

style cluster-alignment loss, and entropy diversity regularization. Sparse top-k routing encourages explicit modality selection, while five-seed geometric-mean ensembling reduces sensitivity to initialization.

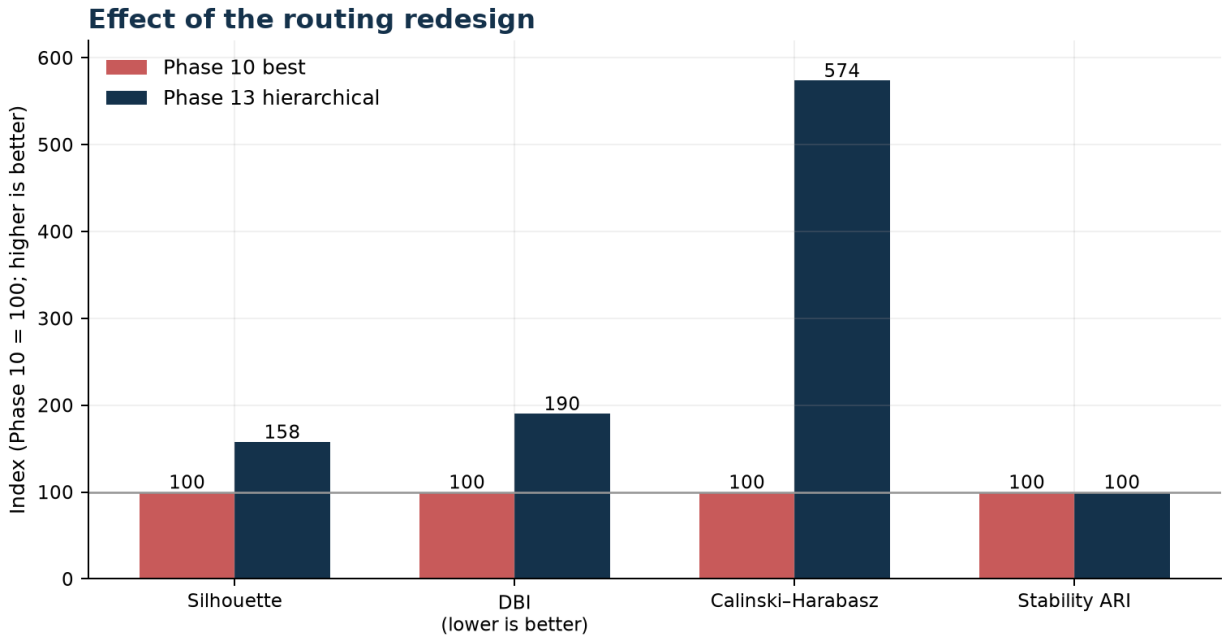


Figure 3: Relative improvement from Phase 10 to Phase 13. DBI is inverted because lower values are better. The resulting hierarchical MoE reaches 0.590873 silhouette, slightly above the static baseline by 0.032945. It also improves DBI from 1.214152 in the best Phase 10 route to 0.638034, increases CH to 19,223.176, and raises stability ARI to 0.999912. The model therefore achieves the desired geometry improvement without sacrificing assignment consistency.

Routing variant	Silhouette	DBI	CH	Stability ARI
Uniform reference	0.288930	1.258587	4,315.472	0.999292
Fixed behavioral-heavy	0.359031	1.259050	3,470.597	0.999397
Cluster-alignment gate	0.349103	1.039966	10,079.353	0.999843
Sparse top-1 routing	0.557691	0.703640	17,210.058	0.999592
Hierarchical MoE	**0.590873**	**0.638034**	**19,223.176**	**0.999912**

7 Retention utility and business interpretation

Retention evaluation highlights a different winner. The KMeans RFM model identifies a maximum-retention cluster at 4.705882%, corresponding to a corrected lift of 3.756427× over the test baseline. The omnibus test is below 0.00001, and the Wilson 95% interval for the maximum cluster is 2.712159%–8.044023%. This makes RFM a powerful, simple benchmark for retention activation.

The initial MoE retention result is weaker: the fixed behavioral-heavy route reaches 2.192982% maximum retention and 1.752785× lift. Phase 13 retention metrics are not yet reported, so its superior geometry should not be treated as proof of superior retention targeting. This is a key decision boundary: deploy Phase 13 for structural segmentation, but evaluate its future retention lift before replacing RFM in lifecycle campaigns.

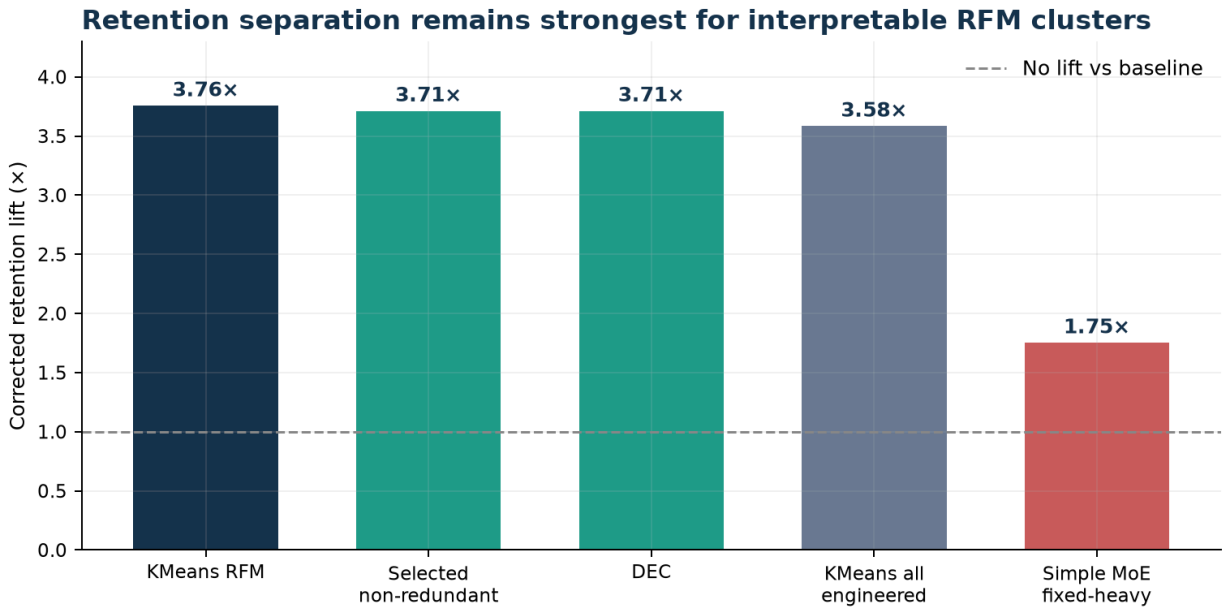


Figure 4: Corrected retention lift reported for models with available downstream evaluation. Phase 13 retention remains an open evaluation item.

8 Reliability controls and limitations

The project includes strong controls against the most consequential failure mode in temporal customer modeling: leakage. Reviews and order events are strictly pre-cutoff; later-known operational fields are excluded; StandardScaler and PCA are fitted on training customers only; validation, rather than test, is used for configuration selection; and future retention labels are reserved for downstream evaluation. All reported leakage checks passed with zero violations.

Several limitations temper the conclusions. First, Phase 6 and Phases 9–13 do not use identical feature sets, dimensionality, or evaluation protocols, so cross-phase comparisons should be read as directional rather than as a single controlled leaderboard. Second, the test set was explored in earlier phases and is not a pristine untouched holdout. Third, retention is rare: only 108 test customers were retained, which widens uncertainty and limits statistical power. Fourth, text coverage is only 39.2%, and text-only comparisons involve a different effective cohort. Finally, 96.88% of customers have frequency one, so many temporal sequences are short and may limit what a sequence encoder can learn.

9 Recommendations

1. **Adopt hierarchical MoE as the leading multimodal segmentation candidate.** Its 0.590873 silhouette, 0.638034 DBI, and 0.999912 stability ARI exceed the static-fusion reference on geometry.
2. **Keep KMeans RFM as the retention control and an interpretable fallback.** Its 3.756× corrected lift is the strongest reported downstream result.
3. **Preserve availability-aware masking.** Explicitly model missing text and renormalize weights; indicator-free fusion materially degrades both geometry and stability.
4. **Monitor gate entropy, sparsity, and modality usage.** These diagnostics are necessary to distinguish genuine adaptive routing from uniform or collapsed behavior.

5. **Run the missing Phase 13 retention evaluation before productionizing campaign decisions.** Geometry leadership is not sufficient evidence for retention uplift.

10 Final assessment

The project successfully demonstrates a conditional form of the adaptive-routing hypothesis. Simple learned gates do not improve segmentation and can substantially underperform a well-designed static baseline. A hierarchical MoE with cluster-alignment loss, sparse routing, entropy regularization, and ensemble averaging does improve geometry, clearing the best static fusion result by 0.032945 silhouette points while delivering the lowest reported DBI among the leading multimodal candidates.

The practical conclusion is therefore not "MoE always wins." It is more precise: **customer-specific routing is valuable when the router is trained to preserve cluster structure, handles missing modalities explicitly, and is evaluated separately for geometry and downstream business utility.** That distinction is the main research contribution of the study.

> **Decision statement.** The hierarchical MoE is the recommended next segmentation model for multimodal analysis; RFM remains the recommended retention benchmark until Phase 13 downstream validation is complete.

Source: project analysis notes supplied for this report · 25 October 2026